AURA - Stage 4

Evidence Model & Logical Reasoning | Technical acceptance report | 24 October 2026

Outcome: The existing AURA verifier now supports bounded compatible-world reasoning over independently validated observations and explicitly authored finite assumptions. It produces the familiar PASS/FAIL/UNCERTAIN certificate plus a new reasoning ledger. This is controlled engineering validation, not external benchmark evidence.

ACCEPTANCE MEASURE	RECORDED RESULT
Automated test suite	115 passing (76 inherited + 39 new Stage 4)
Authored reference scenarios	12/12 expected verdict and status
Fresh controlled SQLite/filesystem workflows	6/6 expected decisions
New schema interfaces	BoundedWorldModel + ReasoningLedger
Stage 5 adaptive inspection	Not yet implemented

1. Architecture and interface

INPUT / MODULE	RESPONSIBILITY
TaskContract + ProbeSpec	Declares every necessary outcome and observable scope
Stage 2 evidence	Captured database/filesystem/trace observations with receipts
Stage 3 admission	Validates source identity, epoch, integrity, fact keys and coverage
Stage 4 finite model	Externally authored domains and safe declarative constraints
Possible-world evaluator	Enumerates evidence-compatible worlds up to a hard cap
Decision and ledger	Conditional verdict, provenance references, world counts, diagnostics

Important: Stage 3 remains a valid conservative baseline. Stage 4 does not import the simulator's hidden actual-world label. Its model assumptions are separate, reviewable inputs.

2. Decision semantics and safety guards

Let K(E,M) contain all assignments inside the finite declared model M that match independently admitted evidence E and satisfy declared model constraints.

VERDICT	CONDITION
PASS	K is nonempty; every compatible world satisfies all obligations and required frame coverage is verified.
FAIL	K is nonempty; every compatible world violates at least one obligation.
UNCERTAIN	Both success and failure remain possible, or decisive evidence/coverage is unavailable.

EDGE CASE	PROTECTIVE RESPONSE
Contradictory independent facts	UNCERTAIN / evidence_conflict
No world matches observations + model	UNCERTAIN / model_inconsistent (never vacuous PASS)
Raw Cartesian domains exceed cap	UNCERTAIN / solver_indeterminate
Frame scope lacks complete receipt	No model-only PASS permitted
All worlds fail, but no single obligation always fails	FAIL / joint_unsatisfiable_completion

A model-relative PASS is conditional on the correctness of the model itself. A deliberately false implication can still force a model-relative PASS, so authored model clauses must never be mistaken for independent proof of external-world effects.

3. Logical findings

Indistinguishable outcomes: The external-notification reference case admits one success candidate and one failure candidate with identical visible evidence. AURA correctly refuses to guess.

Joint failure: A synthetic constraint can establish that complete task success is impossible even when two separate obligations remain unresolved. This is a bounded logical demonstration, not a measured production failure.

Inference sensitivity: The synthetic assumption that a saved report implies a delivered notice turns an unknown case into a conditional model-relative PASS. The assumption is not valid in general and is explicitly excluded from real-world success claims.

4. Controlled evaluation results

NEW CAPTURED SCENARIO	STAGE 3	STAGE 4
Complete workflow	PASS	PASS
Missing report	FAIL	FAIL
Unrelated persistent change	FAIL	FAIL
Misleading success / silent no-op	FAIL	FAIL
Denied independent readback	UNCERTAIN	UNCERTAIN
Skipped mandatory validation	FAIL	FAIL

The full standard evaluation set comprises 12 authored reference inputs + 6 freshly captured local workflows. Stage 4 agrees with Stage 3 in all 18. This is appropriate: without independently justified new model relationships, logical reasoning should not pretend to gain information.

New artifacts

FILE	USE
BoundedWorldModel.schema.json	Finite fact domains, limited formulas, hard enumeration cap
ReasoningLedger.schema.json	Graph, compatible-world counts, witness candidates, hashes, timing
aura_reasoning/model.py	Admitted evidence, world enumeration, contract reasoning
verify_bounded.py	Runs explicit model against saved run or synthetic fixture
reproduce_stage4.py	Regenerates complete local runs, models, ledgers, certificates

Cost accounting

Stage 4 reasoning time is recorded in the reasoning ledger and added to the existing certificate planner-latency cost. Stage 2 capture and inspection costs remain in the total; they are not treated as free preprocessing.

5. Limitations and scientific interpretation

1. Finite candidate worlds are authored abstractions, not the external world. Their completeness and logical constraints must be independently justified.

2. The supplied reference generator can derive domains from observed data; this is valid for software regression but not for blind research evaluation.

3. No third-party agent benchmark, real-service outcome oracle, or systematic cost advantage has been established.

4. No complete monitoring of transient reverted changes or unobserved external APIs has been established.

5. Stage 2 uses a cooperatively quiescent local environment and does not provide a globally atomic multi-store snapshot.

6. Proof by model assumptions does not authenticate causal provenance or guarantee agreement with unstated user intent.

7. The finite enumerator has a conservative hard cap. A general SMT-based solver is not implemented.

8. Stage 5 counterexample-guided probe selection has NOT been built.

6. Reproduction

python -m pip install -e '.[test]'

python -m pytest -q

python reproduce_stage4.py --out ./new_stage4_results

python verify_bounded.py --fixture tests/fixtures/stage1_inputs/05_unobservable_notification --model examples/models/external_uncertainty_reference.json --out ./single_case

All output directories must be new. The archives contain code, schemas, tests, captured evidence and certificates.

7. Next stage

Stage 5 - CERA Adaptive Inspection Planner. Use the evidence graph to select which authorized read-only inspection to execute next, with fixed, greedy, bounded-exact and adaptive comparisons at matched evidence permissions, risk and total acquisition cost. The Stage 4 results do not establish a planning advantage.

GitHub: No commits or pushes performed.